

College of Engineering Talking Points

- Valpo is an “and” institution. Our students study engineering and have other meaningful experiences as well, such as participating in Division I athletics, playing in one of the university’s ensembles, serving at the Chapel, leading a fraternity or sorority, etc. We value and support the well-rounded educational experience that our students seek. The technical skills acquired in the College of Engineering help our students get their first job. The skills that are acquired from a well-rounded general education help our students advance in their careers afterwards. Both are important.
- The Engineering PLUS program is available to students who desire to pair their engineering degree with a second major or a minor. The program allows students to complete summer courses between the first two years and release 9-12 credits from future semesters to be used for the second major or minor.
- We put student learning and success first. Our retention rates (~75% graduate in four years) exceed those at larger institutions by a wide margin. We work very hard to help every student graduate in four years. How?
 - *Teaching-focused faculty who are available outside of scheduled class time.* Faculty in our department are teachers first and are here because they have a passion for helping undergraduate students learn engineering. Faculty make themselves available for students and care about helping them reach their full potential.
 - *Hesse Learning and Resource Center.* The center is staffed by students who receive special training to be effective tutors. When open, students can go for help on assignments and connect with a peer who, ideally, just completed the course themselves. The Hesse Center also organizes study sessions for most of the foundational engineering courses taken in the first two years. The study sessions are led by a Hesse scholar who works closely with the instructor so that the session can provide the support that students need to be successful.
 - *Academic Advising with Faculty.* Professional academic advisors work with students in their first three semesters. Students are then connected with faculty in our department who meet at least once a semester to discuss academic progress and career goals.
 - *Peer Learning.* Other students are often the best resource for learning. Faculty implement active learning during scheduled class time that allows students to work together to solve problems and learn difficult concepts.
- We have a student organization in the college for virtually every interest. Whether it is their particular engineering discipline, a special interest, or their personal identity, there is a group of students who share that passion. These organizations carry out special projects, plan

social events, and arrange professional and educational events that help to develop important skills that complement classroom and laboratory instruction.

- First semester students complete GE 100, Introduction to Engineering. It is a project-based course in which students get their hands on equipment from day one.
 - The first year course is interdisciplinary. All engineering students complete it regardless of their declared major.
 - Projects in GE 100 expose students to all six disciplines of engineering that Valpo offers degrees in. It is easy to change majors after the introductory course.
 - Students learn about the engineering design process in addition to academic skills that will help them to be successful as they matriculate through their program.

- Senior Design
 - All ABET-accredited programs must have a capstone design course or project. What differs between programs is whether the capstone project is interdisciplinary and how long the project is. At Valpo, our senior design course in each department is interdisciplinary and lasts a full year. Interdisciplinary is the reality of the modern workplace, so senior design prepares students for what their work will be like after graduation.
 - Posters in the hallway from previous senior design teams are a great way to showcase the types of projects that students might work on.
 - All projects have a customer who is not part of the team. We strive for every project to have an external, industrial customer.

- Undergraduate Research Opportunities
 - There are lots of opportunities for conducting research with a faculty member.
 - Students can apply up to three credits of research towards their engineering degree through the elective course requirements.
 - Students typically spread research out over three semesters, completing one credit each semester.

Civil and Environmental Engineering Talking Points

- Environmental Engineering Lab (Gellersen 142)
 - Students test surface water samples for different drinking water quality parameters (solids, turbidity, alkalinity, hardness, etc) that are necessary for designing drinking water treat facilities.
 - Students test wastewater samples for parameters necessary for designing wastewater treatment unit operations (e.g. microbial testing, solids, nutrients concentrations, etc.)
 - Students conduct pilot-scale studies related to drinking water treatment processes.
 - Students conduct wastewater treatability studies related to wastewater disinfection processes.
 - The lab is also used for undergraduate student's research.
- Fluids Lab (Fites 186)
 - The fluids lab is an essential part of the junior-level curriculum for civil and environmental engineering.
 - In this lab we measure properties of stationary and moving fluids. These properties are important when:
 - Designing an underwater structure like a dam or submarine because we have to know the forces acting on the structure
 - Designing a bridge so we know how putting bridge piers and/or embankments in the river will affect the flow and water, including its depth and velocity
 - Designing a stormwater management system so we can prevent flooding and improve water quality
 - Designing a water distribution or collection system so we can get clean water to people and dirty water to the treatment plant
- Materials and Structural Engineering Suite (Fites 183)
 - This is where our students get hands-on experience analyzing different construction materials—concrete, steel, and timber. Why is this important? Because different types of concrete and all construction materials serve different purposes
 - I usually ask prospective students to hold two concrete cylinders with different densities (I keep them in the lab) and then ask if they can tell what is different, and how we achieved that difference. The answer is: by changing the ingredients or their proportions.
 - For example, when you drive across a bridge, you trust that the bridge will withstand the weight of your car because it was designed for that purpose. The engineer calculated the required strength and specified the material—whether concrete, steel, or wood—that meets those requirements. But before that material is used, it must be tested in a lab like this one.

- Here, we check material strength (how much load it can withstand), durability (how long it will last before needing replacement), and overall behavior under different loading conditions. To do this, we use equipment such as a universal testing machine (UTS), a compressive strength machine, and a metal impact test machine (Charpy).
- We have a variety of material samples for metals and wood, and we also fabricate our own concrete samples. Students mix concrete here, test different aggregates, cure samples, and then test them.
- Finally, we have a strong floor, a specialized floor designed to resist extreme forces for large-scale structural testing. This facility allows us to test large girders, structural connections, and slabs, and is mostly used for research purposes.
- Soils Lab (Gellersen 141)
 - In the soils lab, the three things we study about soil are:
 - How strong the soil is so that our building doesn't fall over,
 - How compressible the soil is, so that our building settles too much, or worse, unevenly like the Tower of Pisa,
 - How easy or difficult it is to get water to pass through it, which is very important if you want to either put in a well or build an earth dam.
- Transportation Lab (Gellersen 135)
 - The transportation lab focuses on independent undergraduate research, engaging up to five students per year on projects covering the spectrum of transportation: traffic signal operations, validation of traffic signal detectors, microsimulation of traffic flows, pedestrian safety, transportation planning, counting of non-motorized traffic, the use of drones for area mapping, and others.
 - This research tends to be data intensive, hence the number of computers present and the server rack. Being a smaller college, Valpo isn't able to offer an extensive number of electives in transportation so undergraduate research projects give students an opportunity to dig deeper into this area.
 - The lab is also regularly used for WAVES and ASCE meetings, and by students looking for a place to study. It also features an impressive supply of snacks and drinks.

Electrical and Computer Engineering Talking Points

- Relevant to all three large ECE Labs (Gellersen 159, 162, and 166)
 - Hybrid Learning Environment – Combines classroom and laboratory environment in one space, designed specifically for Electrical and Computer Engineering students.
 - Fully Equipped Desks – Each workstation includes essential benchtop equipment:
 - Oscilloscope
 - DC Power Supply
 - Function Generator
 - Digital Multimeter
 - Seamless Integration of Theory and Practice – Instructors can introduce new concepts during lecture, and students can immediately test and apply them at their desks.
 - Reinforced Learning – Students no longer need to wait until a separate lab session to try out what they've learned; application happens in real-time.
 - Active and Engaged Learning – Encourages hands-on experimentation, critical thinking, and immediate feedback during instruction.
 - Industry-Relevant Training – Mirrors the practical setup engineers use in professional environments, preparing students for real-world applications.
- Analog Circuit Laboratory (Gellersen 159)
 - Primarily (although not exclusively) used for teaching courses about analog circuits, such as those involving resistors, capacitors, and inductors. Analog signals can take on a continuous range of values, most typically measured in volts or amps.
- Digital Circuit Laboratory (Gellersen 162)
 - Primarily (although not exclusively) used for teaching courses about digital circuits, such as those that compose computers, in which values can only be 0 or 1. Begins with the study of fundamental digital logic gates like AND, OR, and NOT. Then progresses to the study of Programmable Logic Device and Embedded Microcontrollers, which are small and inexpensive computer chips that are embedded directly into the devices they are designed to control. This is also the room where electrical and computer engineering students typically learn fundamental computer programming skills.
- ECE Makerspace (Gellersen 166)
 - This room is fully equipped with the tools and supplies necessary to fabricate a wide variety of different electronic devices. It is also often used to teach students about electronic circuits, especially those including diodes and transistors, and about both analog and digital communication and control systems.
- ECE Clean Room (Gellersen 157)

- Dedicated Clean Room – A specialized space designed for projects that must remain dust-free.
 - Dust-Proof Environment – Maintains positive air pressure, constantly pushing clean air outward to keep dust and particles from entering.
 - Continuous Air Filtration – Equipped with fine filters that remove even tiny particles, including those naturally produced by people while working inside.
 - Hands-On Space Applications – Currently used by students to build and assemble CubeSats (miniature satellites about $4 \times 4 \times 4$ inches).
 - Real-World Experience – CubeSats are lightweight, low-cost, and increasingly popular because they make satellite launches accessible to smaller companies and organizations.
 - Career-Relevant Training – Students gain direct exposure to the same environment and practices used in aerospace and high-tech industries.
 - Launch to Space – Projects built in the clean room aren't just for the classroom—they're designed to be launched into orbit, connecting learning to real-world impact.
- Wireless Communications Lab (Gellersen 163)
 - Wireless Innovation Lab – A dedicated space where students explore projects in wireless communication, satellite systems, and amateur radio.
 - Professional Soldering Stations – Each station supports circuit assembly from standard through-hole components down to tiny surface-mount devices as small as 2×1.25 mm.
 - Advanced Tools for Precision Work – Students gain hands-on practice using:
 - Microscopes for fine-detail soldering
 - Temperature-controlled soldering & desoldering stations
 - Hot air rework tools for SMD components \ fume extractors for safety
 - Vacuum pickup tools for handling delicate chips
 - Practical Wireless Projects – Students design, build, and test circuits that power wireless transmission, communication links, and radio applications.
 - Satellite Communication Experience – The lab includes a controller system that automatically aligns roof-mounted antennas with passing satellites, enabling students to download real satellite data directly to a campus server.
 - Amateur Radio – Projects extend into amateur (ham) radio and other wireless technologies, giving students real-world exposure to communication systems.
 - ECE Undergraduate Research Lab (Gellersen 164)
 - Provides dedicated space for students and faculty working on an undergraduate research project in electrical or computer engineering.

Mechanical Engineering and Bioengineering Talking Points

- Several Study abroad opportunities are available for students in MEBE:
 - In junior year, students can study in Reutlingen, Germany. Progress towards the engineering degree is not paused. Rather, students can complete three required engineering courses—heat transfer, machine design, and fluid mechanics—while in Germany. Heat transfer is taken from Reutlingen University. Machine Design and fluid mechanics are taken remotely in online courses from Valpo. The online courses are mostly asynchronous, but have synchronous components, e.g. a regular check-in with the instructor and in-person exams administered by the study abroad office at Reutlingen University.
 - Valparaiso International Engineering Program (5 year program)
 - Shorter summer offerings (Ethics or European Infrastructure Course)
- Undergraduate research opportunities are available in many areas of MEBE:
 - Concentrated solar energy
 - Sound and vibration (guitar research)
 - Design optimization
 - Mechatronic systems
 - Next generation manufacturing
 - Bioengineering
- Bioengineering Laboratory (Fites 187)
 - Includes testing equipment that can be used to characterize biological tissues and organs.
 - The large motion-capture field can be used to measure and characterize client movements, allowing recommendations for exercises to improve performance.
 - A wide variety of devices can be used to test and demonstrate the use of bioelectrical signals to actuate and control powered prosthetic devices.
- Rapid Prototyping Laboratory (Gellersen 131)
 - All engineering students have access to the lab on the first day that they step foot on campus and can start using the equipment after completing training sessions. The training sessions have an online component and an in-person component.
 - We look for reasons to say yes, rather than reasons to say no. Students are encouraged to be creative, whether it's for class, a student organization, or a personal project.
 - New printer acquisitions ensure that we have the latest and best technology that students may see for prototyping on the job.
 - Bambu Labs H2D printers. These printers were released in 2025 and our department acquired some of the first to be released in the USA. The printers

have two print heads and can print in two different styles of plastic at once. These units allow us to print designs with relative motion. Examples are on the table (pink interlocking sheet and blue expanding star).

- Bambu Labs X1E printers. These are enterprise grade printers that are built for businesses. Our X1E printers have the capability of multi-color prints with up to 16 colors.
- Bambu Labs A1 mini. This portable printer allows us to bring printing on the road and students have brought the unit to local schools to share engineering and inspire the next generation.
- Carbon fiber laser (yellow) and CO2 laser (blue) enable creative designs. The lasers can etch or cut through certain materials. Examples of what can be created include printed circuit boards and water bottles with names etched on the outside.
- Manufacturing Lab (Gellersen 170)
 - Required laboratory course in the 2nd year that teaches conventional machining operations that are completed on lathes, mills, and drill presses.
 - Students learn how to operate the machines because it is the best way to learn their limitations. Knowing limitations makes our students better design engineers and provides the foundation for expanding knowledge in manufacturing engineering.
 - Students draw parts and then make them. They keep what they make. Many alumni still proudly display parts made in the laboratory course (e.g. sheet metal box, pawn piece, parallel clamp)
 - Two CNC mills and one CNC lathe.
 - Wazer Waterjet cutter - material agnostic. Precise control and clean cut surfaces.
 - Plasma Torch on a gantry system - material must be conductive. Cuts through ¼” steel like its butter.
- Mechatronics Lab (Gellersen 134)
 - Exposes students to modern control and instrumentation systems.
 - Supports the Mechatronics minor. Many laboratory courses for the minor are taught in this space.
 - Students learn how to program the ABB robot sitting in front of the window. The task is to pick up colored chips rolling down the conveyor belt and sort them into cups based on color.
 - Two required laboratory courses are supported in this lab.
- Energy Systems Suite (Fites 186)
 - ME students have two required laboratory courses supported in the lab.
 - Some experiments that could be highlighted include depending on prospective student interest:

- Wind tunnel. Measure lift and drag on objects and airfoils and compare with engineering model predictions.
 - Two-stage air compressor with intercooling. Measure the energy required to compress the air and the importance of cooling the compressed air between the two compressor stages.
 - Axial flow fan. Teaches blower performance curves. Measure energy use, pressure rise, and flow rate.
 - Centrifugal pump. The backbone of processes involved liquid flow. Students study pump curves and predict flow and pressure rise. Measurements are made and compared with an engineering model.
 - Peloton Wheel Impulse Turbine. Small scale model of turbines common in the northwestern United States. The peloton wheel uses a jet of water blasted into cups. Students compare an engineering model of efficiency against measurement.
 - Converging-diverging nozzle. This is a benchtop model of a rocket nozzle. Gas is accelerated to supersonic speeds inside of the nozzle. Students compare measurements of the flow rate to an engineering model prediction. Students also visualize the normal shock that forms in the diverging section of the nozzle.
 - Diesel engine. A 4-cylinder Perkins engine mounted on a test stand. An eddy current dynamometer is used to control the load on the engine and measure its output.
 - 25-kW micro-gas turbine. The unit burns natural gas to drive an electric generator. When running, the electricity is put into the building. The unit was supposed to go into the World Trade Center before September 1. Students measure how the rpm of the engine influences the power output and efficiency.
 - Renewable Energy Zone. Small wind turbine and 3 PV panels on a two-axis tracker.
- Materials and Structural Engineering Suite (Fites 183)
 - Students have a required materials and mechanics laboratory that is supported in this lab. Students use the tensile testers, the cold-rolling machine, hardness testers, and microscopes to look at metal grains, and high-temperature ovens to study the impact of heat treating on mechanical properties.
 - Find a broken dog bone sample to show. They are on the laboratory table to the left of the black cabinets.
- Solar Energy Research Facility (SERF Building)
 - This facility does not study solar panels that produce electricity. Rather, the focus of the teaching and research is on the use of concentrated solar energy as a source of heat

for high-temperature processes. The overarching objectives are to study ways to use the high-temperature heat to produce fuels from water, carbon dioxide, and/or biomass or to produce high-value commodities like Magnesium or Zinc.

- The cornerstone instrument of the facility is the solar furnace. The solar furnace is a small-scale concentrating device that emulates the concentration that one can achieve in larger scale systems.
- Concentration is achieved in two steps. In the first, the heliostat (mirror structure to the north of the building) is controlled in two axes (up-and-down and east-and-west) to direct sunlight into the building. The redirected sunlight is not yet concentrated. In the second step, the light strikes the concentrator. The concentrator is made up of 303 hexagonal mirror facets.
- Each facet outputs light in a cone (focuses), and the facets are arranged such that the tip of each cone comes together at the focal point. This point is about 10 ft in the air above the test stand. The test stand is used to lift equipment and place it into the focal point.
- The facility supports an elective course in energy.
- The facility also supports undergraduate research in solar energy.